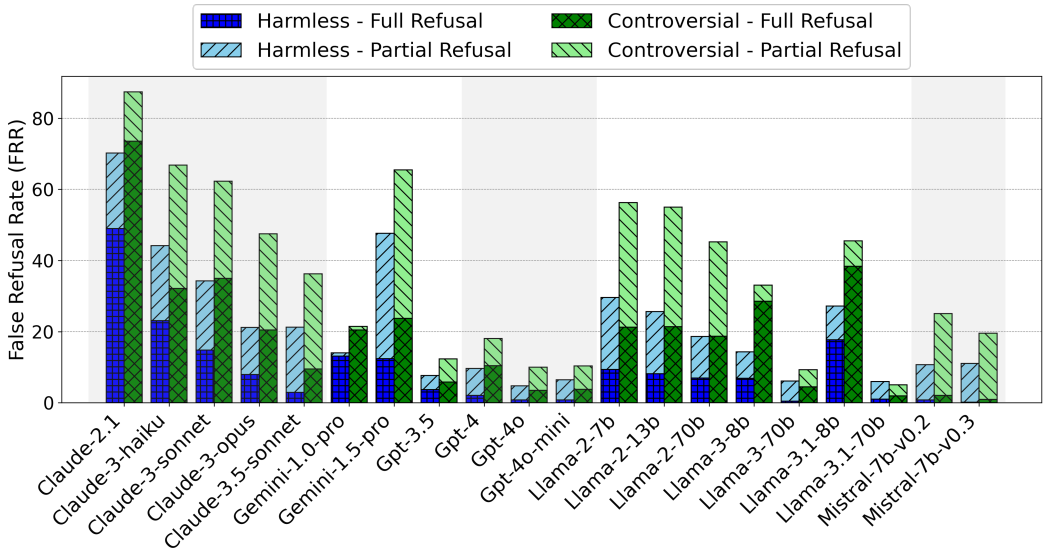

Figure 6: False refusal rates of different LLMs on PHTest.

real-world use scenarios of LLMs. If needed, our method can also generate specifically distributed prompts through content steering to reflect domain-specific scenarios.

5 Evaluation

This section uses PHTest to evaluate the false refusals of popular LLMs. Our test models include GPT-3.5, 4, 4o, 4o-mini (OpenAI, 2023), Claude-2.1, 3 (Haiku, Sonnet, Opus), 3.5 (Sonnet) (Anthropic, 2024), Llama3-Instruct (8B, 70B), 3.1 Instruct (8B, 70B) (Dubey et al., 2024), 2-Chat (7B, 13B, 70B) (Touvron et al., 2023), Gemini-1.0-pro, 1.5-pro (Reid et al., 2024), Mistral-7B-Instruct-V0.2 and v0.3 (Jiang et al., 2023). We use greedy decoding (zero temperature) for consistent results.

Following Röttger et al. (2023b), we categorize model responses into three cases: **(1) Full refusal**, where the model declares refusal and does not subsequently answer the user's request. **(2) Partial refusal**, where the model initially refuses but then answers the user's request. **(3) Full compliance**, where the model does not declare refusal. We prompt GPT-4 to label the model response into one of these three categories. We measure false refusal rates (FRRs, %), and abbreviate false refusal prompts as PHPs.

5.1 Results

Figure 6 shows our evaluation results. Overall, the FRRs of the different models vary significantly, with the Claude and Llama2 families showing notably higher FRRs compared to others. Although more capable models do not necessarily show lower FRRs, for models within the same family (potentially undergone similar alignment processes), larger ones tend to have lower FRRs than smaller ones. Furthermore, our dataset yields the following exclusive conclusions:

PHTest reveals new insights due to fine-grained labeling and scale. Results on XSTest (Figure 3 in Anthropic (2024)) show that Claude 3 Haiku and Sonnet have a false refusal rate similar to Claude 2.1, indicating no improvement in reducing false refusals. However, results on our dataset show a moderate decline on controversial PHPs (from 87% to 69%, 62%) and a significant drop on harmless PHPs (from 70% to 44%, 34%) for Haiku and Sonnet compared to Claude 2.1. This suggests that Claude 3 is better at identifying (clearly) harmless pseudo-harmful requests but still faces limitations due to developers' risk preferences on controversial requests.

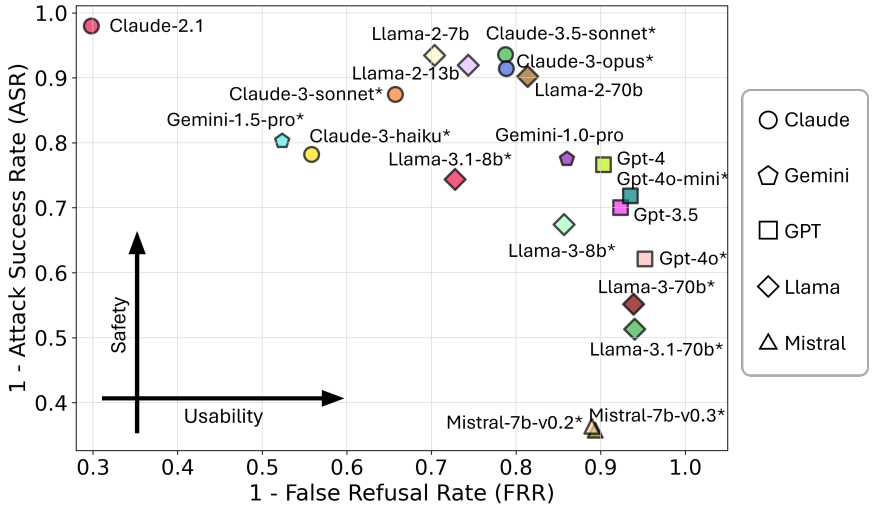

Figure 7: Tested LLMs demonstrate a trade-off between safety (low ASR on HarmBench) and usability (low FRR on PHTest's harmless prompts). The safety of *-marked LLMs are potentially underestimated. We test their jailbreak ASR on a small available prompt set from HarmBench, while taking others directly from HarmBench's report.

Model size vs false refusal. Figure 6 shows that scaling up the model size reduces FRRs. For example, enlarging Claude 3 from Haiku to Opus decreases FRRs from 44% to 21% on harmless PHPs and 66% to 47% on controversial PHPs; enlarging Llama 3.1 from 8b to 70b reduces FRRs from 27% to 6%, and 45% to 5% on harmless and controversial PHPs respectively. However, this does not mean that training larger models alone can solve the problem of false refusals. Figure 7 shows that larger models (e.g., Llama 3.1 70b compared to 8b) sometimes have lower FRRs because they compromise on safety.

New models have reduced false refusals, but not always. Figure 6 shows that Claude and Llama, which previously struggled with frequent false refusals, have greatly reduced false refusals in their latest models. However, models like Gemini 1.5 Pro have seen an increase in false refusals in their newest updates.

5.2 Safety vs False-Refusal Trade-off

We further evaluate the trade-off between LLM's safety and false refusal. For this task, Röttger et al. (2023b) test safety using a set of blatantly harmful prompts from XSTest, resulting in GPT-4's nearly perfect trade-offs. Here, we instead test safety on jailbreak prompts (Mazeika et al., 2024) that, contrary to pseudo-harmful prompts, use various strategies to disguise harmful requests, thus better reflecting the model's safety performance in malicious user scenarios. We cite results from Mazeika et al. (2024) for models' safety performance under jailbreak attacks and re-evaluate the missing ones.

Figure 7 shows the trade-off between safety and usability across various LLMs, with none clearly dominating the others. GPTs strike a moderate balance, while Claude 2.1 achieves the highest safety at the cost of the highest FRR. This trade-off may partly result from the lack of comprehensive pseudo-harmful prompts used as negative samples during safety alignment, making it harder to finetune the model's refusal boundaries. Our method, however, can generate model-specific pseudo-harmful data to help mitigate this trade-off.

5.3 Jailbreak Defenses Should Be Calibrated by False-Refusal Rates

While most jailbreak defense works do evaluate their impact on usability, these evaluations often focus on open conversations (e.g., WildChat Zhao et al. (2024b)) or specific tasks (e.g.,

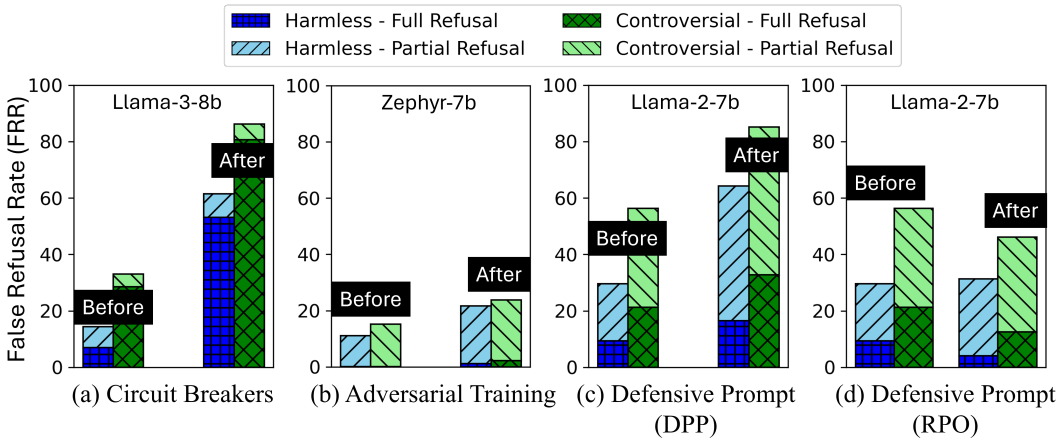

Figure 8: False refusal rates before and after applying some jailbreak defenses.

math or coding problems) that rarely trigger false refusals. Since false refusals are sporadic (yet with serious consequences), these tasks cannot reliably reflect the real-world impact on usability. Therefore, we test four jailbreak defenses on PHTest, including Circuit Breakers (Zou et al., 2024), adversarial training (Mazeika et al., 2024), defensive prompts-DPP (Xiong et al., 2024), and defensive prompts-RPO (Zhou et al., 2024). Figure 8 reveals that the Circuit Breaker and DPP considerably raise FRRs, whereas RPO maintains it. Since false refusal rates directly hinder developers' efforts to make LLMs safer during finetuning, we advocate that all defense methods should also evaluate their impact on usability using PHTest and other pseudo-harmful prompt datasets.

6 Related Work

Safety alignment of LLMs. The development cycle of large language models (LLMs) includes multiple stages of safety alignment to ensure their behavior aligns with human values (e.g., see Llama3's technical report Dubey et al. (2024)). During pre-training, developers filter data to exclude harmful content, reducing the likelihood of the model generating such content. In fine-tuning, developers use supervised fine-tuning and RLHF (Ouyang et al., 2022; Bai et al., 2022; Dai et al., 2023; Rafailov et al., 2024) with safety-related positive and negative examples (e.g., BeaverTails Ji et al. (2024)) to adjust the model's refusal boundaries. Finally, at deployment, system-level safety filters (e.g., Llama Guard Inan et al. (2023) and ShieldGemma Zeng et al. (2024a)) detect and block harmful inputs or outputs. A side effect of such alignment — often called the "alignment tax" — is that LLMs sometimes overfit simple rules in training data (Stephan et al., 2024), leading to false refusals.

Safety-Usability Trade-off. The trade-off between safety and usability (aka. harmlessness and helpfulness) in language models has been a long-standing issue. Earlier work focuses on how safety alignment affects LLM performance on general tasks like factual QA, mathematical reasoning, and coding (Bai et al., 2022; Ganguli et al., 2022; Chen et al., 2023). Recent studies discuss the false refusal issue, also referred to as exaggerated safety, over-defensiveness, and overkill. Bianchi et al. (2023) find that too much safety-tuning makes models refuse perfectly safe prompts if they superficially resemble unsafe ones. Varshney et al. (2023) find that self-checking-based jailbreak defenses, which prompt the LLM to check its own input and output, significantly increases false refusal rates on some harmless prompt datasets. Some LLMs' technical reports also discuss the false refusal behaviors (Anthropic, 2024; Dubey et al., 2024; Inan et al., 2023).

Some other work aims to either mitigate or exacerbate this trade-off. Based on the interpretability of refusal behaviors (Zou et al., 2023a; Arditi et al., 2024), Cao et al. (2024) identify the refusal vectors in LLM's representation spaces and steer them to strike a better trade-off, whereas Stickland et al. (2024) take an additional step before steering to avoid side effects. Zhao et al. (2024a) use post safety alignment to mitigate the trade-off. On the other hand,